



MECH MAINT- BOILER Department
SIMHADRI



Ref: SMTTP/MECH MAINT-/2022-23/LMI/602862

Date:

17/11/2022(08:53:48)

Subject: APPROVAL OF LMI BOILER HEADER LIGAMENT CRACKING -
GUIDANCE ON INSPECTION POLICY

OPERATION & MAINTENANCE

MECHANICAL MAINTENANCE

(PRESSURE PARTS)

REF.: SMPP/O&M/MM/PP/

Date: 17.11.2022

Subject: Review of LMI “BOILER HEADER LIGAMENT CRACKING-GUIDANCE ON INSPECTION POLICY”

LMI No: LMI/OGN/OPS/MECH/004

The above approved LMI was revised in the month of July.’2022. The envisaged changes in LMI are incorporated as per details at revision index of LMI.

Put up for approval please.

Submitted by:

राहुल जैन - वरिष्ठ प्रबंधक

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On:17/11/2022(08:53:48) **Ref:**SMTTP/MECH MAINT-/2022-23/LMI/602862

May please provide approval for the revised version of subject LMI document with rev-5.

Submitted by:

सुरेश बाबु कोल्ला - उप महाप्रबंधक

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On:19/11/2022(20:06:46) **Ref:**SMTTPP/MECH MAINT-/2022-23/LMI/602862

Changes incorporated as per revised LMI:

1. Eco Header figure added
2. Eco header included in low risk header
3. Inspection of uncracked header elaborated
4. Annexure-A checklist added for Boiler Header ligament cracking

The revised LMI (Doc- New LMI attached) may please be approved.

Submitted by:

ओम प्रकाश - अपर महाप्रबंधक

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On:23/11/2022(09:14:42) **Ref:**SMTTPP/MECH MAINT-/2022-23/LMI/602862

Submitted for approval pls

Submitted by:

झिमलि धर - अपर महाप्रबंधक

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On:29/11/2022(09:32:14) **Ref:**SMTTPP/MECH MAINT-/2022-23/LMI/602862

Revised LMI, may please be accorded approval.

Submitted by:

मैथ्यू ई कोवूर - महाप्रबंधक

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On:29/11/2022(10:53:06) **Ref:**SMTTPP/MECH MAINT-/2022-23/LMI/602862

कृपया अनुमोदन प्रदान किया जाए।

May be approved please.

Submitted by:

राजेश कुमार झा - महाप्रबंधक

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On:29/11/2022(11:45:44) **Ref:**SMTTPP/MECH MAINT-/2022-23/LMI/602862

Approved & Submitted by:

गिरीश चन्द्र चौकसे - मुख्य महाप्रबंधक

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On:30/11/2022(12:34:34) **Ref:**SMTTPP/MECH MAINT-/2022-23/LMI/602862

LOCATION MANAGEMENT INSTRUCTION

LMI/OGN/OPS/MECH/004/

ISSUE NO.1; REVISION NO. 5.00; DATE: NOV'2022

**BOILER HEADER LIGAMENT
CRACKING - GUIDANCE ON
INSPECTION POLICY**

APPROVED FOR

***Implementation*-----**

HOP, SIMHADRI

Date-----

LMI APPROVAL DOCUMENTS

<u>1.0 TITLE OF LMI</u>	BOILER HEADER LIGAMENT CRACKING - GUIDANCE ON INSPECTION POLICY
<u>2.a) DOCUMENT CODE</u>	<u>LMI/OGN/OPS/MECH/004/</u>
b) REVISION/ISSUE NO.&DATE	Rev. No.: 5, November 2022
<u>3.ORIGINAL/REF.DOCUMENT</u>	<u>OGN/OPS/MECH/004</u>
<u>4.PREPARED BY:</u> Sr. MANAGER (MM-PP)	SIGNATURE-----
<u>5. CHECKED BY:</u> DGM(BMD) AGM (BMD) GM (MTC.) GM (O&M)	SIGNATURE_____ SIGNATURE_____ SIGNATURE_____ SIGNATURE-----_____
<u>6. REVIEWED AND APPROVED BY</u>	SIGNATURE_____ HOP, SIMHADRI
<u>7.DATE OF ISSUE</u>	
<u>8.INDEX LIST</u>	ENCLOSED.
<u>9.DISTRIBUTION LIST</u>	ENCLOSED.

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<u>1.0 INDEX</u>		
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<u>2.0 INTRODUCTION</u> The need to operate fossil power plants in a flexible manner has resulted in the appearance of thermal fatigue damage in the components such as steam headers. Because the cost of replacing such components can be high and lead times for their manufacture long, there is an increased need for an inspection strategy that can locate and monitor flaw growth so that informed decisions can be made on the incidence of maintenance expenditure. The objective of this LMI is to establish an inspection strategy for ligament crack inspection of headers so that informed decisions can be made with respect to operation and maintenance. <u>3.0 SCOPE</u> The scope of this LMI is to detail the methods and periodicity that are to be adopted for the header ligament crack inspection of steam headers. Steam headers are divided into two categories (a) High risk (i) All super heater stage headers (i.e., the collecting header at the outlet of a superheater stage of tubing) operating above drum saturation temperature and (ii) all headers in horizontal self-draining super heaters which operate above 400 degree C (b) Low risk i.e., all super heater, economiser and reheater steam headers not identified in the High-Risk category. <u>4.0 REFERENCE DOCUMENT</u> Operation guidance note OGN/OPS/MECH/004 circulated by operation services-NTPC corporate center. <u>5.0 SUPERSEDED DOCUMENT</u> BOILER HEADER LIGAMENT CRACKING – GUIDANCE ON INSPECTION POLICY REV 2.00 DATE:APR'2013 <u>6.0 SPECIFIC INSTRUCTIONS</u> To achieve the objective of this LMI the following approach is to be followed: a) Periodic Inspection of header ligaments for cracks. b) Remedial Actions to sustain continued operation on discovery of internal cracks c) Reinspection of low-risk headers only if cracking detected in previous inspection. DPT/MPI of all stubs of economizer inlet header in overhauling.		

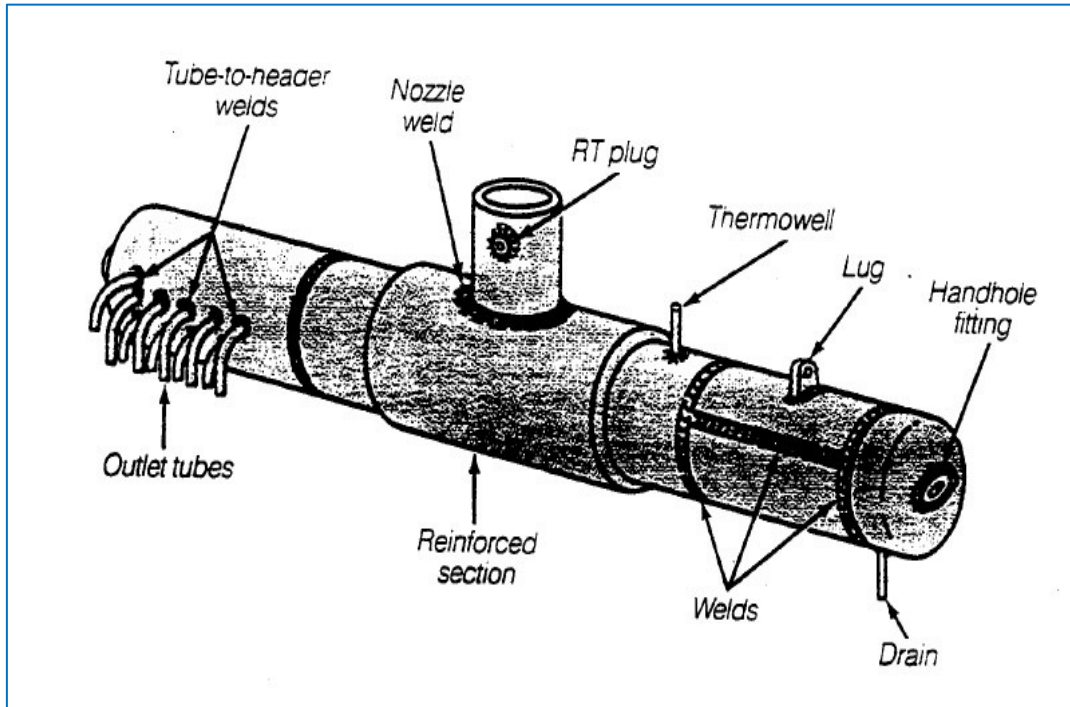
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<u>7.0 METHODOLOGY OF IMPLEMENTATION AND RESPONSIBILITY CENTERS</u> <i><u>I) Inspection procedure to be adopted</u></i> A) During overhauls i) visual inspection of surface for any cracks ii) LPT at suspected locations to determine the size of the surface crack and recording the same to facilitate crack growth monitoring. B) During RLA studies i) An initial internal visual inspection of header to be carried out using boroscope. ii) if cracking is present, an ultrasonic thickness inspection shall be carried out to determine the crack sizes and profiles RESPONSIBILITY : HOS (Pr .Parts group) <i><u>II) Inspection strategy to be adopted</u></i> a) <i>For uncracked high-risk headers :</i> - IA) during every overhaul - IB) after 10 years of operation - IC) Reinspected at or before twice the number of starts as at first inspection. IOT/MPI/DPT to be done for terminal/antlers tubes every overhauling. b) <i>For cracked high risk headers where crack growth is to be monitored :</i> - IA) during every overhaul - IB) every two years c) <i>For replaced high risk headers of same material as original with or without geometry improvements :</i> - IA) during every overhaul - IB) at half the number of starts of the life of original header d) <i>For high-risk headers replaced with 9% Cr Steel header:</i> - IA) during every overhaul - IB) at twice the number of starts of the life of original header e) <i>For low-risk headers:</i> - IA) during every overhaul - IB) after 10 years of operation - RESPONSIBILITY : HOS (Pr. Parts group)		

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<u>III)Actions in case an internal crack is discovered</u> a) provide a safety case around the header, if deemed necessary b) replace complete header by using same material as original or modified 9% Cr steel c) replace a part of header on recommendation of OEM. - RESPONSIBILITY : HOS (Pr .Parts group)												
<u>8. EXCEPTION REPORTING</u> The exceptions (internal ligament cracks) encountered during inspection are to be reported as per the following format and to be taken up with corporate engg & manufacturer ----- HOD (BMD) <table border="1"> <thead> <tr> <th>Description of exception</th> <th>Unit No.</th> <th>Header description</th> <th>Location of exception</th> <th>Remarks</th> </tr> </thead> <tbody> <tr> <td> </td> <td> </td> <td> </td> <td> </td> <td> </td> </tr> </tbody> </table>			Description of exception	Unit No.	Header description	Location of exception	Remarks					
Description of exception	Unit No.	Header description	Location of exception	Remarks								
<u>9.0 RESPONSIBILITIES</u> The responsibility of carrying out different tasks in relation to inspection and maintenance of integrity of steam headers as detailed in the LMI are with HOD (BMD).												
<u>10. RECORD KEEPING</u> The records to be maintained under this LMI include a) compliance record of the activities detailed under this LMI b) reports of survey and rectification work carried out c) Exception reporting The above records are to be maintained HOD (BMD).												
<u>11.REVIEW</u> This document shall be reviewed by HOP on TWO yearly basis.												

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12.0 REVISION INDEX (for changes/revision)				
Sl. No.	Rev. no.	Date	Details	Approved by
01	1.00	12.04.10	i) Approval routing changed at "LMI APPROVAL DOCUMENT"	
			ii) Responsibility changed from SS (PP) to DGM (PP) at responsibilities	
			iii) Reviewing authority changed from DGM (MM) to GM I/c	
02	2.00	30.04.13	i) Responsibility changed from DGM (PP) to HOS (Pressure Parts) at Responsibilities	
			ii) Inclusion of U#3 & 4 in compliance record	
			iii) Distribution List	
03	3.00	10.01.2020	i) Approval routing changed at "LMI Approval document"	
			ii) Review Frequency changed from "three years" to "TWO years"	
			iii) Distribution List	
04	4.00	01.07.2020	i) Paragraph 3(a) High Risk (i) has been modified as per paragraph 5.1 (a) of COS-ISO-00-OGN/OPS/MECH/004 Rev. No.: 1 May 2020	
			ii) Distribution List	
05	4.00	05.07.2022	No changes done	
06	5.00	16.11.2022	i)Changes done in bold as per COS-ISO-00-OGN/OPS/MECH/004 Rev 2, Sept 2022	

[illegible]

NTPC	14.0 COMPLIANCE RECORD	SIMHADRI
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UNIT NO.	ACTIVITY	COMPLIANCE
I/II/III/IV		



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Annexure A Checklist for Boiler Header Ligament Cracking

S.No.	Activity	Frequency	Document	Done (Yes/ No)
1	All tube stubs of Economiser inlet header checked by DPT/MPI	Overhauling	Protocol	
2	All High-risk headers drains, and vents stub weld checked by DPT/ MPI	Overhauling/ Opportunity shutdown	Protocol	
3	Insitu Oxide Scale thickness (IOT) measurement done in Terminal/ Antlers tubes of SH/ RH headers	Overhauling/ Opportunity Shutdown	IOT readings	
4	Terminal/ Antlers tubes having highest IOT sent to NETRA for assessment of its remaining life	Overhauling/ Opportunity Shutdown	NETRA Report	
5	Header bore holes and header ligaments inspected by DPT/ Boroscope for tube having highest IOT and its adjacent area/ header bore hole in SH and RH headers.	Overhauling/ Opportunity Shutdown	Protocol	
6	Header bore holes and header ligaments inspected by DPT/ Boroscope for tube having stub cracks in Economiser inlet header	Overhauling/ Opportunity Shutdown	Protocol	
7	Headers Stub cracks reported to Corporate OS	Immediate after crack detection	Protocol	